SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25

Question Count: 5

Code of Conduct

I __J.P.BADRI(192524301)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: ___ J.P.B _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.

   How can encryption and access control protect sensitive backup data?

   Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Auto-Scaling and Load Balancing

Cloud auto-scaling automatically increases or decreases computing resources based on traffic demand. Load balancing distributes incoming requests across multiple servers to prevent overload.

- Auto-scaling: Adds servers during seasonal sales and removes extra servers when demand decreases.
- Load balancing: Distributes requests evenly across available servers.
- Reduced downtime: Prevents server overload and redirects traffic away from unhealthy instances.
- Monitoring tools: Track CPU usage, memory, request rates, and response times to trigger scaling actions.
- Real-world example: An e-commerce website such as Amazon can use auto-scaling and load balancing to handle increased traffic during festive sales.

Conclusion: These techniques improve availability, reduce response time, and provide a better user experience while controlling costs.

2. Secure Cloud Backup Storage

Cloud storage services provide durable and redundant storage to protect backup data against hardware failures and disasters.

- Durability: Multiple copies and integrity checks help prevent permanent data loss.
- Redundancy: Data can be replicated across storage devices or availability zones.
- Encryption: Encryption at rest and in transit protects sensitive backup data.
- Access control: IAM roles, strong authentication, and least-privilege permissions prevent unauthorized access.
- Cost advantages: Pay-as-you-go pricing and archival storage reduce hardware, maintenance, and electricity costs.
- Disaster recovery example: If a company's local server fails, it can restore its database and application backups from cloud storage.

Conclusion: Cloud backup provides reliable recovery, improved data protection, and cost-effective long-term storage.

3. Content Delivery Networks (CDNs)

A CDN is a distributed network of edge servers that delivers content from locations closer to users.

- Reduced latency: Content travels a shorter distance from nearby edge servers.
- Caching: Frequently requested images, videos, and files are stored at edge locations.
- Improved performance: Reduces load on the origin server and speeds up content delivery.
- Global availability: Cloud providers offer geographically distributed regions and multiple availability zones.
- Reliability: Traffic can be redirected when an edge server or regional endpoint becomes unavailable.
- Real-world example: A global gaming platform can use a CDN to deliver game updates and media efficiently to players in India, Europe, and America.

Conclusion: CDNs improve loading speed, reduce latency, and deliver a more consistent experience for users worldwide.

4. Cloud Security and Regulatory Compliance

Financial institutions must protect sensitive information and comply with applicable financial and data-protection regulations.

- Regulatory compliance: Cloud providers offer security controls, compliance documentation, and certifications to support audits.
- Logging: Records user activity, account access, and financial transactions for investigation.
- Monitoring: Detects suspicious activity and generates security alerts.
- Encryption: Protects sensitive information both at rest and in transit.
- Access control: Multi-factor authentication and least-privilege policies restrict access to authorized personnel.
- Data sovereignty: Select approved geographic regions, verify where data and backups are stored, and enforce contractual and regulatory requirements.
- Example: A bank can encrypt transaction records, log every database access, and monitor unusual transfers to identify potential fraud.

Conclusion: Strong access controls, encryption, continuous monitoring, and regular audits improve security and help meet compliance requirements.

5. Containers and Virtual Machines

Feature	Containers	Virtual Machines
Architecture	Share the host OS kernel	Run separate guest operating systems
Performance	Lightweight with low overhead	Generally require more resources
Startup	Usually starts quickly	Usually takes longer to start
Resource usage	Efficient CPU and memory usage	Higher resource requirements
Isolation	Process-level isolation	Hypervisor-based isolation
Portability	Portable with compatible container runtimes	Portable with compatible hypervisors and VM formats

Role of Kubernetes:

- Automates container deployment and scheduling.
- Scales application replicas based on demand.
- Restarts failed containers and supports service discovery.
- Enables rolling updates with reduced disruption.

Portability: Containers package applications and dependencies together, helping teams deploy consistently across development systems and different cloud providers.

Real-world example: An online shopping application can use separate microservices for user accounts, product catalogues, payments, and orders. Kubernetes manages and scales each service independently.